

Kestrel headlamp assembly condensation and LED module failure

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Area: bulletins

1 Condition

Kestrel vehicles may exhibit internal condensation within the front headlamp assembly, part number HLMP-7020, progressing to intermittent or complete failure of the integrated LED module. Operators report fogging on the inner lens after rain or car wash exposure, followed by flickering low-beam output. Inspect the HLMP-7020 housing seal and breather vent before any replacement decision. Lighting circuit context is provided in the Kestrel Body Service Manual :: Lighting Sub-System .

2 Cause

The condensation results from a compromised housing-to-lens seal on the HLMP-7020 assembly, which permits moisture ingress that corrodes the LED driver board and shorts the module. Once moisture reaches the driver, the LED elements fail progressively and cannot be serviced separately. Body electrical fundamentals relevant to this fault are outlined in the Kestrel Body Service Manual :: Body Electrical System Overview . Note that a comparable sealed-assembly design is identified in the Ridgeline HD Body Service Manual :: Headlamp Assembly Identification .

3 Repair Procedure

Replace the affected headlamp assembly HLMP-7020 as a complete unit following procedure PROC-HEADLAMP, ensuring the new housing seal and breather are seated correctly to prevent recurrence. Perform an aim verification and a moisture-soak check after installation. The complete steps are documented in the Kestrel Body Service Manual :: Headlamp Assembly Replacement Procedure , with an equivalent cross-platform procedure in the Ridgeline HD Body Service Manual :: Headlamp Assembly Replacement Procedure . For unrelated emissions complaints on this platform, see Kestrel 1.5T: Catalyst efficiency below threshold from degraded converter .